



Michigan Tech

Course: Clinical decision Support and AI Modelling

Course Code: SAT 5141

Course Instructor: Professor Guy Hembroff

Project Title:

Personalized Risk-to-timeline modelling for cervical cancer progression:

Predicting outcomes from clinical and population risk factors.

Group Members

Valeria Mudzindiko

Lusubilo Nyondo

Abstract

Cervical cancer remains a leading global health challenge despite advances in screening and vaccination. While recent studies have demonstrated promising diagnostic frameworks making use of machine learning for early detection of cervical cancer (Shakil et al., 2024; Yang et al., 2019), less attention has been directed towards modelling disease progression in a way that supports personalized clinical decision-making. This gap is mostly seen in low- and middle-income countries, where delayed detection and limited follow-up resources worsens differences in outcomes (Ahishakiye et al., 2020).

Our research investigates Personalized Risk-to-Timeline Modelling for Cervical Cancer Progression, using a [dataset](#) of 858 patient records containing 36 demographic, behavioral, and clinical risk factors. While prior work emphasizes on binary classification of presence or absence of disease (Shakil et al., 2024; Yang et al., 2019), our approach seeks to learn patient-specific risk trajectories over time, incorporating both clinical test outcomes (e.g., Biopsy, Cytology, Schiller, Hinselmann) and population risk determinants (e.g., age, sexual history, contraceptive use, and STDs). The study aims to predict the likelihood of cervical intraepithelial neoplasia (CIN) or cancer and the estimated time-to-event of progression by integrating temporal modelling with explainable AI to recommend actionable timelines for screening, referral, or intervention.

The methodological framework builds upon peer-reviewed work on ensemble learning, feature selection, and interpretable models such as SHAP, while adapting these advances for a personalized progression context. Expected outcomes include the development of a decision-support system that is both predictive and transparent, enabling clinicians to prioritize high-risk patients and tailor follow-up intervals. Our project seeks to advance the shift from population-level detection to individualized trajectory-based decision support, thereby contributing to precision prevention and equitable cancer care.

References

Ahishakiye, E., Wario, R.D., Mwangi, W. and Taremwa, D., 2020. Prediction of Cervical Cancer Basing on Risk Factors using Ensemble Learning. In: IST-Africa 2020 Conference Proceedings. IST-Africa Institute and IIMC. <https://www.researchgate.net/publication/341204214>

Shakil, R., Islam, S. and Akter, B., 2024. A precise machine learning model: Detecting cervical cancer using feature selection and explainable AI. Journal of Pathology Informatics, 15, p.100398. <https://doi.org/10.1016/j.jpi.2024.100398> .

Yang, W., Gou, X., Xu, T., Yi, X. and Jiang, M., 2019. Cervical Cancer Risk Prediction Model and Analysis of Risk Factors based on Machine Learning. In: Proceedings of the 2019 International Conference on Bioinformatics and Biomedical Technology (ICBBT). ACM, pp.50-55. <https://doi.org/10.1145/3340074.3340078>.

Link to dataset: <https://archive.ics.uci.edu/dataset/383/cervical+cancer+risk+factors>